

## Smart grid: concept and status

Smart grid combines power system infrastructure and communication infrastructure to create a power network with enhanced efficiency, reliability, safety and better utilisation of fast depleting natural resources. Rather than requiring a new power grid, it involves automated monitoring and control of the existing power infrastructure, ensuring uninterrupted power supply to consumers while reducing losses and increasing productivity.

Across the globe, remote monitoring and control power systems are being favoured to achieve benefits such as lower operating and maintenance costs, lesser peak demand, reduction in power theft, increased revenue, reliability and quality.

The government of India too has taken cognizance of this trend and is taking steps towards a smarter grid. Its accelerated power development and reforms programme (AP-DRP) provides funds to state electricity utilities to improve energy auditing and accounting systems in order to reduce losses by 15 per cent.

The Smart Grid Task Force set up by the government is an inter-ministerial group that creates a roadmap with the objective of accelerating the smart grid developments in India.

Consumers stand to gain from smart grid implementation in many ways. A fully automated distribution system will allow power utility companies to introduce flexible tariff systems, enabling consumers to select an electricity consumption pattern and reduce their electricity bills.

Power Grid Corporation of India Limited is implementing a smart grid test bed at Pudukcherry in partnership with public and private players. It has signed MoU with as many as 60 companies.

— Prof. (Mrs) Mini Shaji Thomas, Member, IEEE Educational Activities Board, Chair, Section outreach committee (SEOC), and Professor & CPIO, Jamia Millia Islamia, New Delhi

portfolio in the field of FACTS and in-house manufacturing of key components.

FACTS devices are classified as shunt compensation devices (static VAR compensators, static compensators, etc), series compensation devices (thyristor-controlled series capacitors, static series synchronous compensators, etc) and a combination of series and shunt compensation devices (unified power flow controllers).

Static VAR compensation systems help compensate reactive power in dynamic mode, thus helping better regulation on transmission line, reducing over-voltage and transmission line losses.

Prof. Thomas informs, "In India PGCIL has installed many thyristor-controlled series capacitors and static VAR compensators in transmission systems. Distribution companies are using extensive capacitor compensation with thyristor control.

"High-voltage direct current (HVDC) transmission is another application of power electronic devices in transmission of electric power. HVDC lines are used for transmitting a large quantity of power over long distances, at 550 kV and above voltages.

"India has many HVDC lines and PGCIL is a world leader in this technology implementation. There are also HVDC back-to-back stations to connect two asynchronous systems and to enable controlled transfer of power."

## Balancing load and generation

The smart grid concept tries to achieve a balance between load and generation. With increased focus on renewable sources of energy such as wind, solar and hydro, this imbalance is likely to increase. Hence a smart grid application is important to help store electric energy. Whenever the load is in excess of generation, energy can be tapped from these storage devices and thus load demand managed.

"The government of India has identified 20 pilot projects where this concept is to be implemented before scale-up. In its full scope, the concept also covers smart devices at home that can be preprogrammed to 'switch on, switch off' based on the power scenario. For example, in a dairy or a large air-conditioning plant, the chiller can be switched on when there is surplus power at night, helping meet daytime peak demand," informs Venu.



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